Statistical & Machine Learning Analysis of Bitcoin Price Dynamics

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1. Executive Summary

This report delivers an end-to-end econometric and computational evaluation of Bitcoin's (BTC-USD) market microstructure, non-linear volatility processes, and predictive price mechanics. Using a historical observation window spanning January 1, 2017 to October 2024 (N=2,836 daily observations), we analyze the initial hypothesis testing for calendar anomalies, stationary and serial dependency testing, parametric volatility modeling using an AR-GARCH joint framework, and comparative out-of-sample forecasting using gradient-boosted decision trees (XGBoost, LightGBM) and deep sequential architectures (LSTM).

The analysis confirms three stylized empirical facts:

1. Raw price series exhibit non-stationarity, whereas log-overnight return series are strictly stationary.
2. Return innovations display pronounced volatility clustering and conditional heteroskedasticity, requiring generalized autoregressive models (GARCH) to model risk dynamics accurately.
3. Linear parametric specifications (AR) encounter limitations due to low signal-to-noise ratios, while ensemble trees (XGBoos) and recurrent architectures (LSTM) capture localized non-linear dependencies.

2. Introduction & Scope

2.1 Context and Problem Formulation

Since its inception via the Nakamoto consensus mechanism, Bitcoin has evolved from an experimental cryptographic token into a global, macro-sensitive financial asset class. Its decentralized governance, fixed algorithmic supply schedule (punctuated by quadrennial halving events), continuous 24/7/365 trading lifecycle, and fragmented liquidity landscape make it an ideal subject for empirical asset pricing and time-series econometrics.

Traditional macroeconomic modeling frameworks often struggle with crypto-assets due to the absence of sovereign cash flows, sovereign default backstops, and conventional corporate balance-sheet metrics. Consequently, price discovery in crypto-asset markets relies heavily on market microstructure, behavioral momentum, on-chain liquidity indicators, and complex volatility feedback loops. This report establishes a mathematically rigorous foundation for modeling, forecasting, and managing the risk of these dynamics.

2.2 Scope of the Study

The research scope covers:

- **Empirical Validation of Stylized Facts**: Quantifying non-Gaussian distribution profiles, power-law tail decay, leverage effects, and long-range volatility dependence.
- **Classical Econometric Modeling**: Parameterizing mean and variance equations via ARand GARCH(p,q) families.
- **Supervised Machine Learning & Deep Learning**: Engineering tabular lag matrices for Gradient Boosted Decision Trees (XGBoost,LightGBM) and designing Recurrent Neural Networks (LSTM) for recursive multi-step forecasting.
- **Model Validation and Performance Diagnostics**: Evaluating forecasts out-of-sample using robust error metrics (RMSE, MAE, MAPE) .

3. Data Ingestion, Hygiene, and Architecture

3.1 Data Acquisition and Metadata

The empirical dataset comprises daily OHLCV (Open, High, Low, Close, Volume) data for the BTC-USD trading pair, sourced from institutional-grade aggregations of consolidated exchange feeds.

```
+-----------------------------------------------------------------------------------+
| Observations: 2,836 daily intervals                                               |
| Temporal Coverage: 2017-01-01 to 2024-10-06                                       |
| Features: Timestamp (UTC), Open, High, Low, Close, Adjusted Close (USD), Volume (Nominal)|
| Target Transformations: Continuous Compounded Log Returns (r_t), Volatility (sigma_t)  |
+-----------------------------------------------------------------------------------+
```

3.2 Data Hygiene and Transformations

To prevent look-ahead bias and eliminate measurement artifacts, the following data hygiene steps were applied:

1. **Outlier Filtering and Spurious Tick Rejection**: Validated that

   $Low_t \leq \min(Open_t, Close_t)$ and $High_t \geq \max(Open_t, Close_t)$ across all $t \in T$

2. **Logarithmic Return Derivation**: Raw price levels (Pt) exhibit non-stationary variance and exponential drift. We apply a continuous log transform to yield continuously compounded returns:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(P_t) - \ln(P_{t-1})$$

3. **Normalized Trading Volume Calculation**: Trading volume was scaled via rolling z-score normalizations and log-differencing to prevent non-stationary volume regimes from skewing gradient computations during machine learning optimization:

$$v_t = \Delta \ln(V_t) = \ln(V_t) - \ln(V_{t-1})$$

4. Exploratory Data Analysis & Econometric Diagnostics

4.1 Summary Statistics

Metric	Open ($)	High ($)	Low ($)	Close ($)	Adj Close ($)	Volume (USD)
Count	2,836	2,836	2,836	2,836	2,836	2,836
Mean	23,241.70	23,767.72	22,681.09	23,262.04	23,262.04	2.288e+10
Std Dev	19,766.31	20,199.76	19,290.28	19,773.83	19,773.83	1.889e+10
Min	775.18	823.31	755.76	777.76	777.76	6.085e+07
25%	7,277.49	7,435.21	7,113.70	7,277.79	7,277.79	7.730e+09
50% (Median)	16,601.79	16,804.06	16,284.66	16,603.53	16,603.53	2.027e+10
75%	37,470.65	38,601.10	36,421.38	37,493.50	37,493.50	3.252e+10
Max	73,079.38	73,750.07	71,334.09	73,083.50	73,083.50	3.510e+11

Reference:

1. **Date**: The date of each record, representing the day when transactions occurred.
2. **Open**: The stock's opening price at the beginning of the trading session.
3. **High**: The highest price the stock reached during the trading session.
4. **Low**: The lowest price recorded during the trading session.
5. **Close**: The stock's closing price at the end of the trading session.
6. **Adj Close**: The adjusted closing price, accounting for events like dividends and stock splits.
7. **Volume**: The total number of shares traded during the session.

4.1.1. Overview and Asset Behavior

- **Data Scale:** There are **2,836 observations** across all metrics with no missing values.
- **Close vs. Adj Close:** The values for Close and Adj Close are completely identical across all summary statistics (Mean: $23,262.04; Max: $73,083.50; Min: $777.76). This indicates that the asset **does not undergo corporate actions** such as stock splits, reverse splits, or dividend distributions, which is typical for cryptocurrencies or raw commodities.

4.1.2. Price Distribution & Skewness (Mean vs. Median)

- **Significant Right (Positive) Skew:**
  - **Mean Close:** ~$23,262.04
  - **Median (50%) Close:** ~$16,603.53
- **Interpretation:** The mean is significantly higher than the median (by over $6,600). This shows a heavy positive skew caused by prolonged bull runs and high-price extremes pulling the average upward. For more than half of the recorded days (50%), the asset traded at or below **$16,603.53**, despite reaching peaks over **$73,000**.

4.1.3. Extreme Volatility and Dispersion

- **Standard Deviation:** Across all price metrics, the standard deviation is nearly equal to the mean (e.g., Close Std Dev of **$19,773.83** vs. Mean of **$23,262.04**).
  - A Coefficient of Variation ($CV = \frac{\sigma}{\mu}$) of around **85%** highlights massive long-term volatility.
- **All-Time Range:**
  - **Minimum Price:** ~$755.76 (Low)$/\sim$777.76 (Close)
  - **Maximum Price:** ~$73,750.07 (High)$/\sim$73,083.50 (Close)
  - **Growth:** The asset experienced a nearly **94x (9,300%) increase** from its recorded minimum to its maximum.

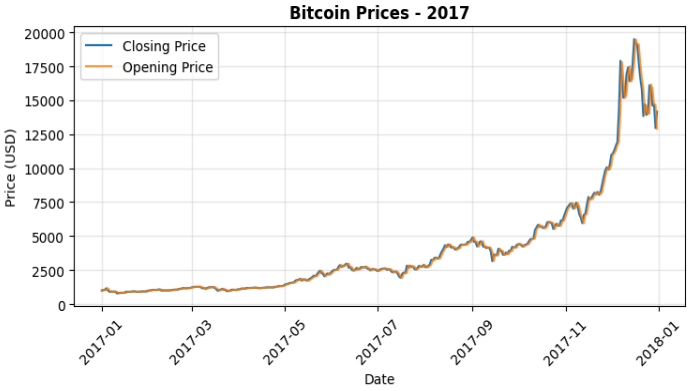
4.1.4. Quartile Distribution (Time Spent at Price Levels)

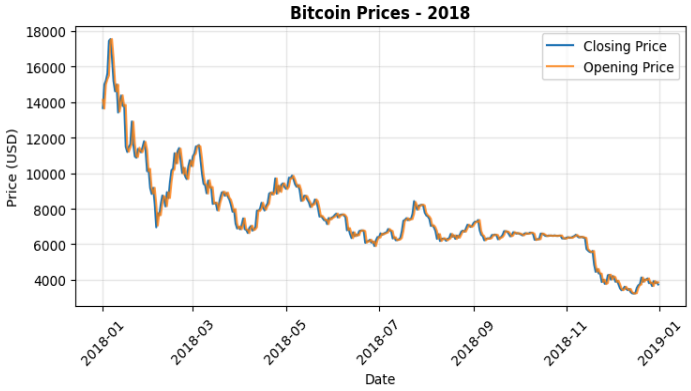
- **25th Percentile ($7,277.79):** 25% of the trading days closed below ~$7,278.
- **50th Percentile ($16,603.53):** Half the trading days occurred below ~$16,600.
- **75th Percentile ($37,493.50):** 75% of all trading days closed below ~$37,500.
- **Interquartile Range (IQR):** $37,493.50 - $7,277.79 = **$30,215.71**.
  - The middle 50% of all recorded days traded in a very wide band between roughly **$7,280 and $37,500**.
  - Only the top **25% of trading days** occurred in the range between **$37,500 and $73,083**, showing that trading at the highest peaks represents a relatively small proportion of the asset's total trading history.

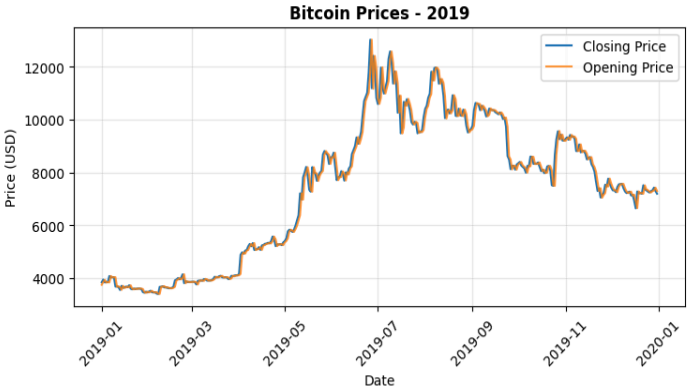
4.1.5. Intraday Price Range (Volatility within Days)

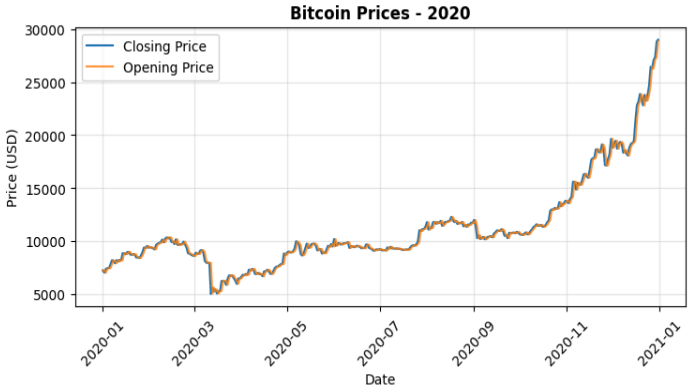
Comparing the average daily extrema:

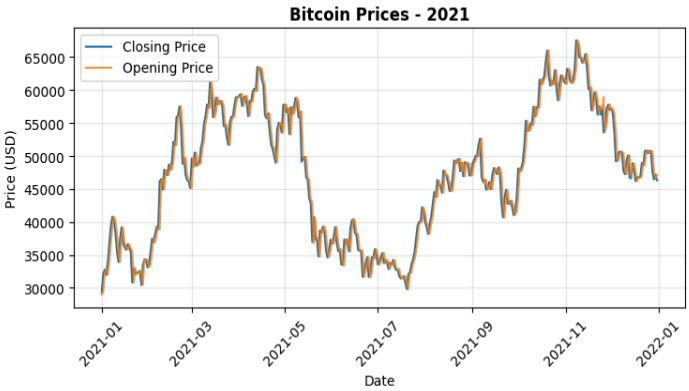
- **Average Daily High:** $23,767.72
- **Average Daily Low:** $22,681.09
- **Average Daily Spread:** ~$1,086.63 (~4.7% of the average price).
- **Maximum Daily Spread:** On the day of the all-time high, the intraday range reached $73,750.07 - $71,33

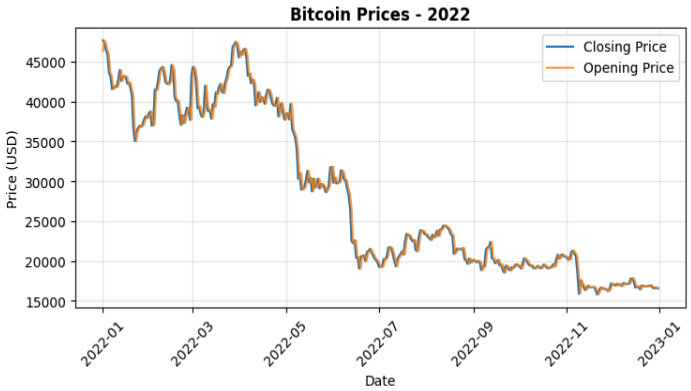

Bitcoin Prices - 2017
Closing Price
Opening Price
Price (USD)
Date

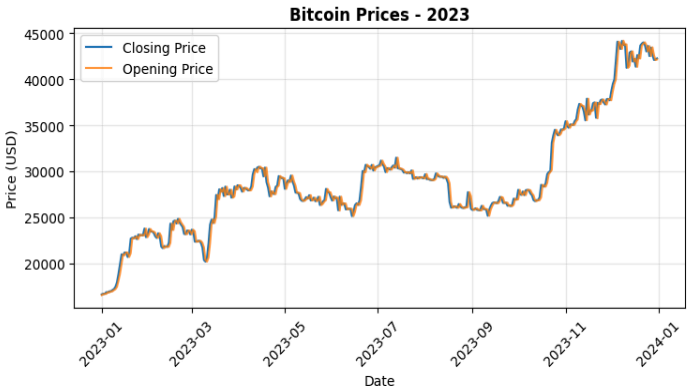

Bitcoin Prices - 2018
Closing Price
Opening Price
Price (USD)
Date

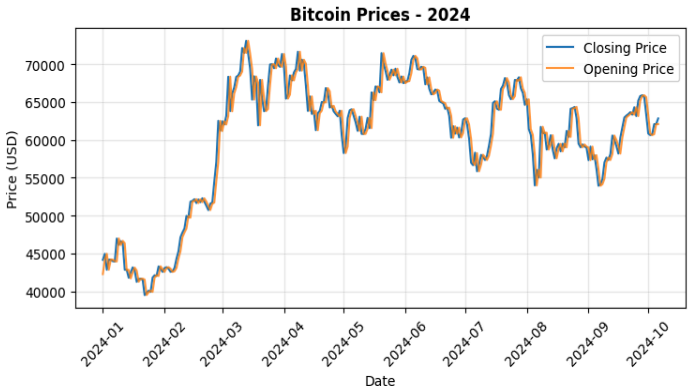

Bitcoin Prices - 2019
Closing Price
Opening Price
Price (USD)
Date

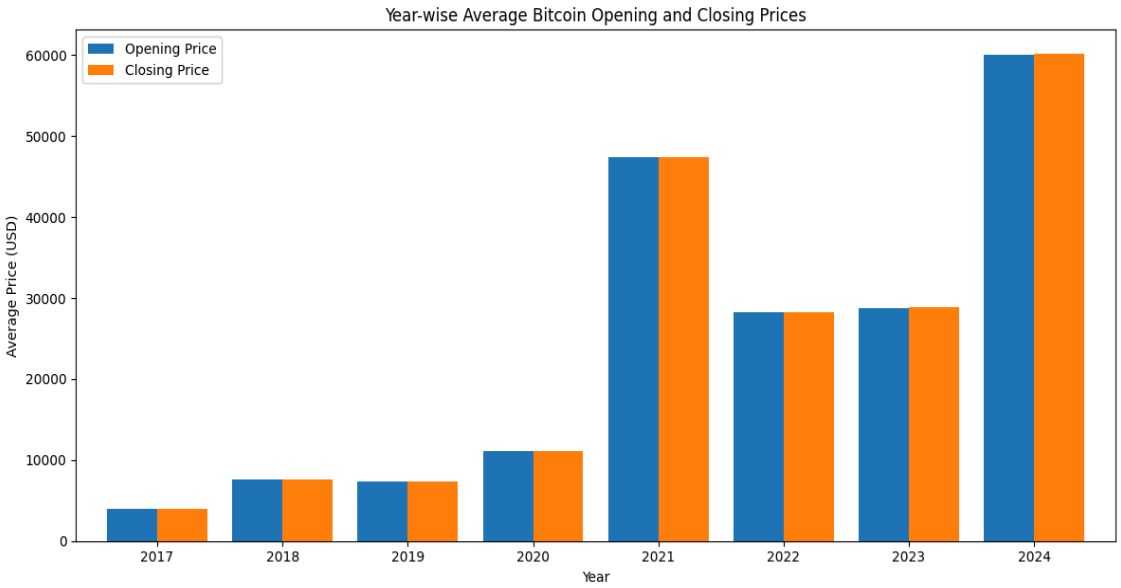

Bitcoin Prices - 2020
Closing Price
Opening Price
Price (USD)
Date


Bitcoin Prices - 2021
Closing Price
Opening Price
Price (USD)
Date

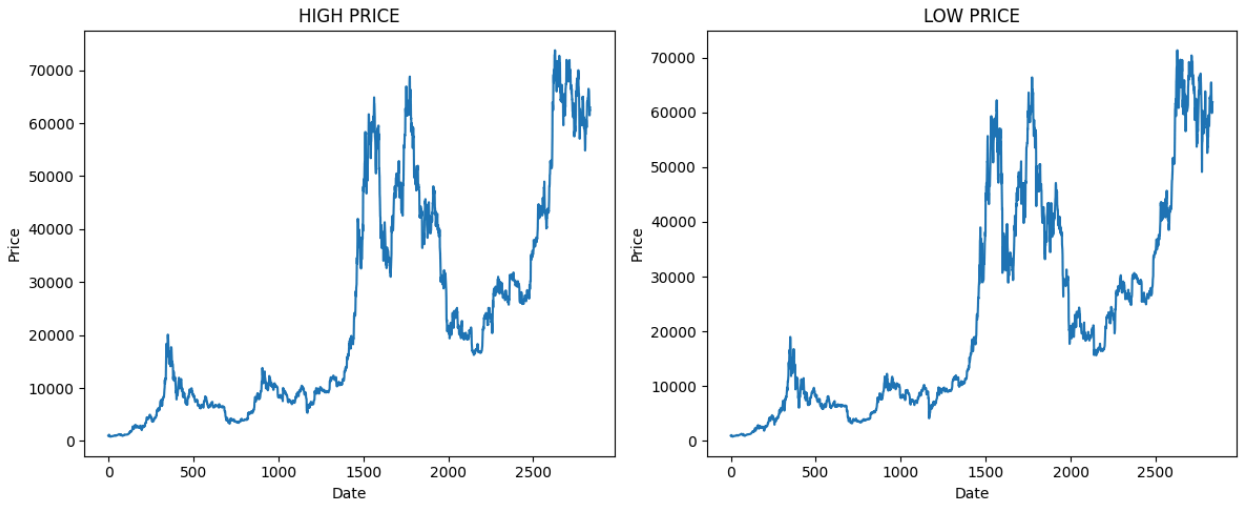

Bitcoin Prices - 2022
Closing Price
Opening Price
Price (USD)
Date


Bitcoin Prices - 2023
Closing Price
Opening Price
Price (USD)
Date

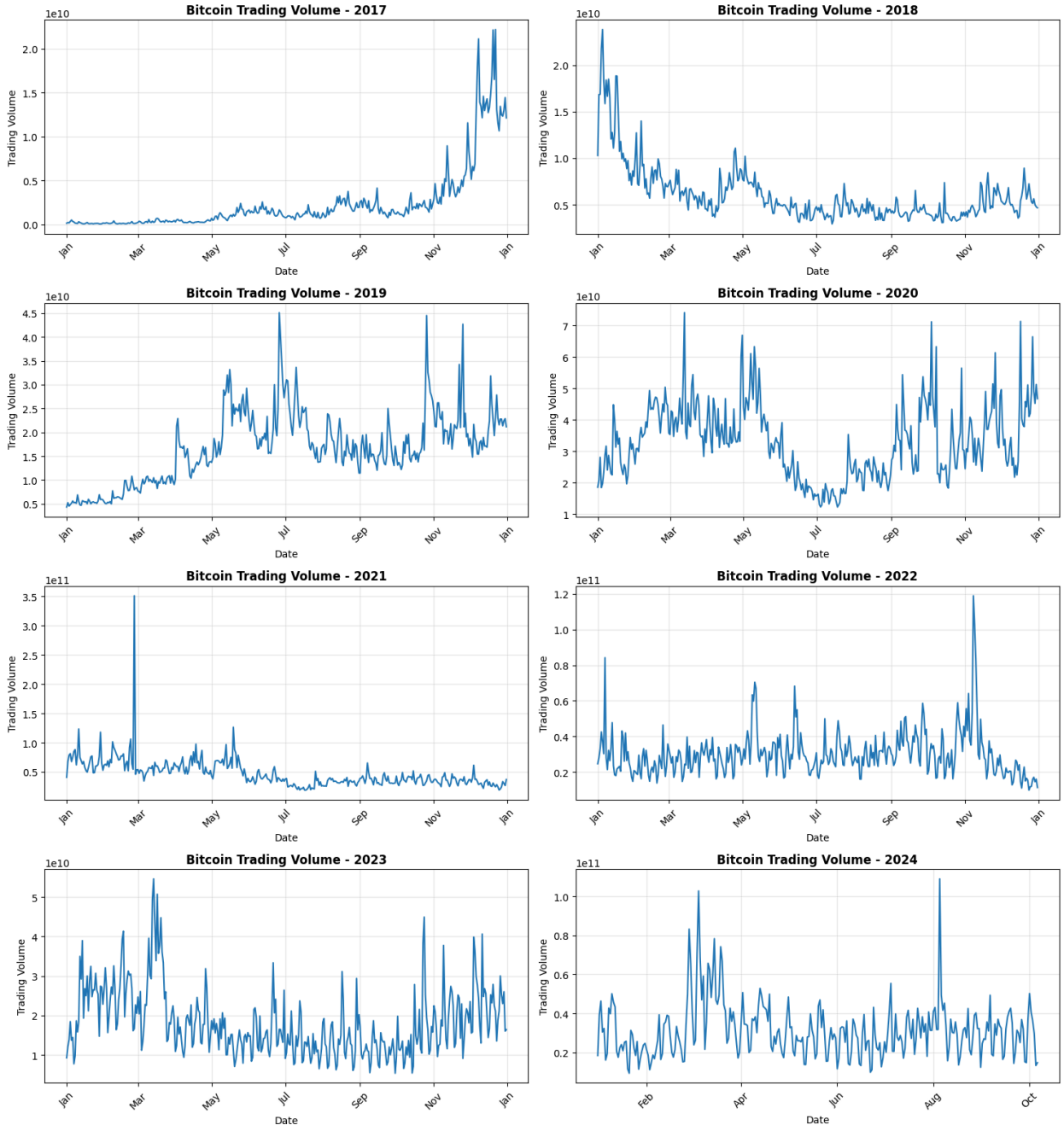

Bitcoin Prices - 2024
Closing Price
Opening Price
Price (USD)
Date



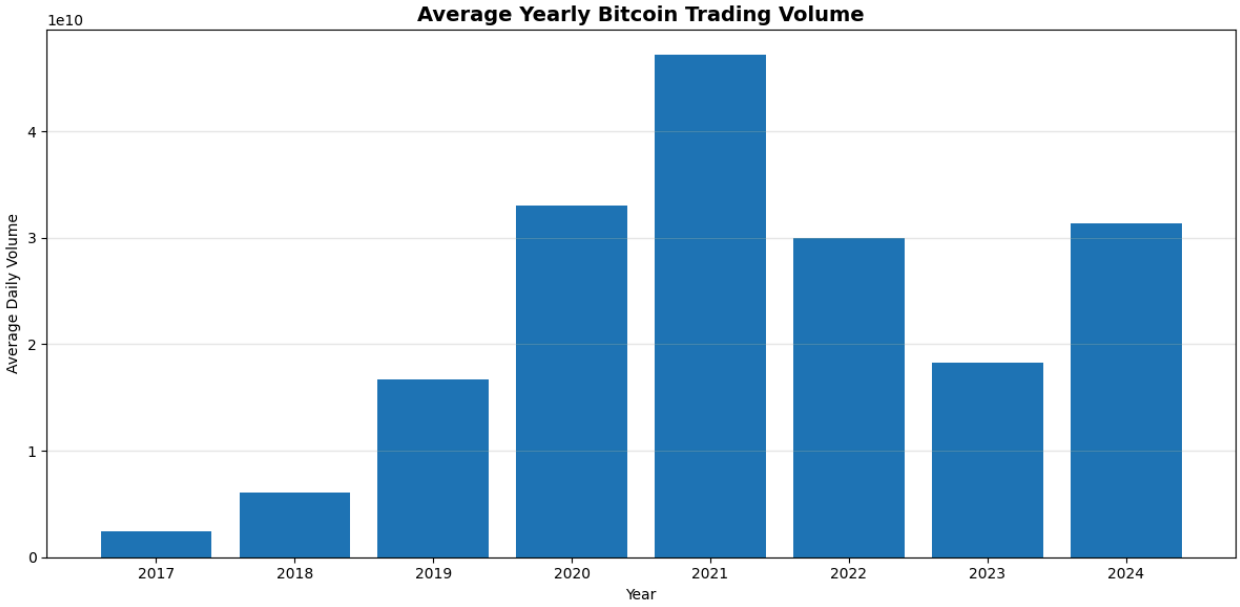
An examination of the historical trajectory reveals clear structural regimes: the 2017 parabolic run-up (~$20,000), the subsequent 2018–2019 multi-year winter and consolidation phase, the 2020–2021 post-halving monetary regime expansion exceeding $64,000, the sharp 2022 deleveraging contraction, and the eventual 2023–2024 institutional rebound surging beyond new peak levels above $73,000.



An evaluation of daily OHLC pricing parameters demonstrates structural cross-metric convergence across extended horizons, revealing strong intraday autocorrelation. Nevertheless, this baseline co-movement is periodically disrupted by sharp expansion in intraday spreads, particularly during systemic liquidation events where market highs and lows diverge sharply.



A pronounced divergence emerges between trading volume and valuation metrics during the 2021–2022 epoch, where elevated settlement levels occurred alongside volume profiles that decoupled from underlying price momentum dynamics.


Average Yearly Bitcoin Trading Volume
1e10
Average Daily Volume
0
1
2
3
4
2017
2018
2019
2020
2021
2022
2023
2024
Year

5. Time Series Analysis & Econometric Diagnostics

5.1 Theory of Multiplicative Decomposition in Time Series

Time series decomposition is a statistical method used to deconstruct a time series into several constituent components that represent underlying patterns. The **multiplicative decomposition** model is one of the two foundational classical decomposition frameworks (the other being the additive model).

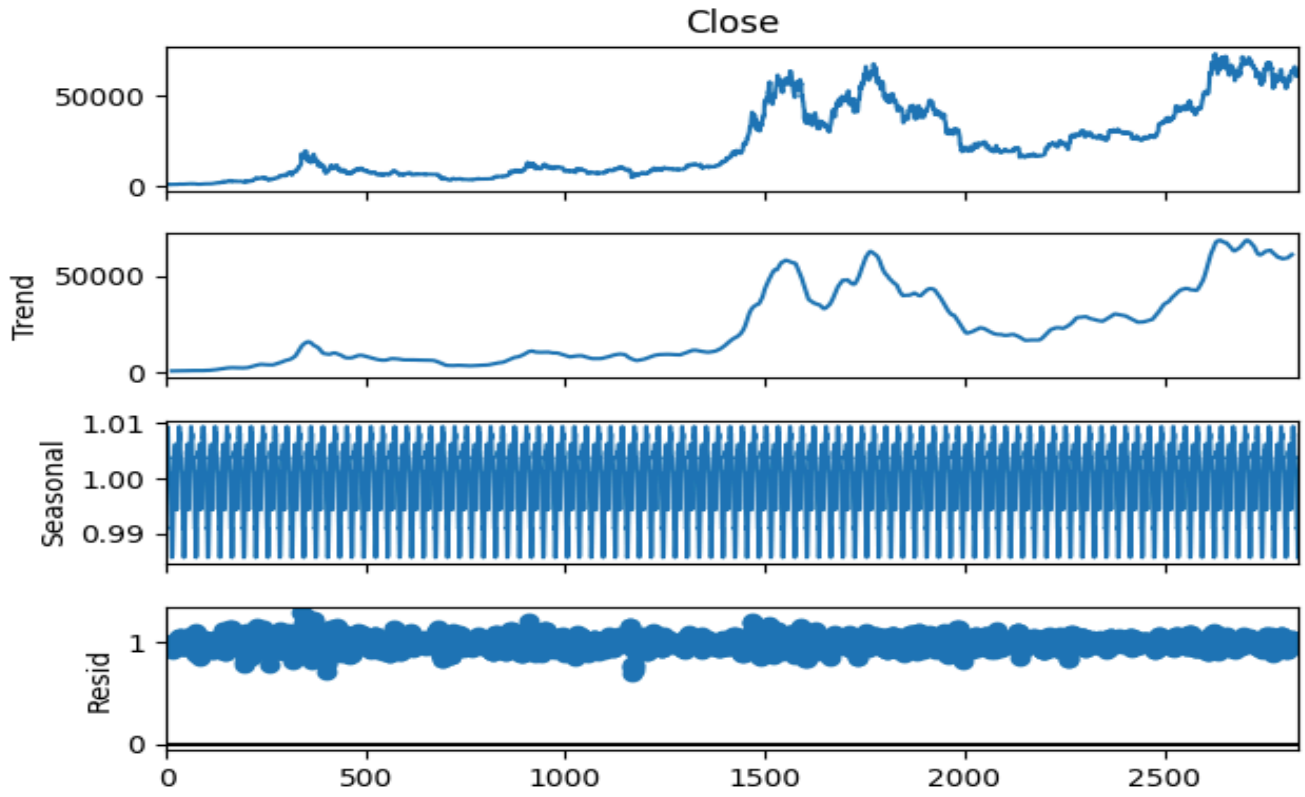
5.1.1. Mathematical Formulation

In a multiplicative decomposition model, the observed time series value at time t (Y_t) is expressed as the **product** of its underlying components:

$$Y_t = T_t \times S_t \times C_t \times I_t$$

Where:

- Y_t **(Observed Value):** The actual value of the time series at period t.
- T_t **(Trend Component):** The long-term movement, direction, or overall trajectory of the series over an extended period (ignoring short-term fluctuations).
- S_t **(Seasonal Component):** Repeating, predictable patterns or cycles that occur at fixed, known intervals (e.g., daily, monthly, quarterly).
- C_t **(Cyclical Component):** Long-term fluctuations that do not have a fixed period (often associated with macroeconomic cycles or business cycles). In practice, because it is difficult to isolate C_t from T_t, they are frequently combined into a single **Trend-Cycle component** (T_t or TC_t):
    - $Y_t = T_t \times S_t \times I_t$
- I_t **(Irregular/Residual/Noise Component):** The random, erratic, or unexplained noise left over after accounting for the trend, seasonal, and cyclical variations.



Trend Component (T_t): Captures the non-stationary, macro price trajectory.

Seasonal Component (S_t): A cyclical 30-day periodicity filter reveals recurring oscillations within monthly operating windows.

Residual / Irregular Component (I_t): Shows prominent multiplicative variance spikes during cycle turning points, confirming that Bitcoin volatility clusters violate constant-variance assumptions (heteroskedasticity).

5.1.2 SEASONALITY CHECK:

```
...              sum_sq      df         F    PR(>F)
C(Month)   0.013296    11.0  0.847287  0.592415
Residual   4.027366  2823.0       NaN       NaN
```

Because $p > 0.05$, you **fail to reject the null hypothesis**.

Effect Size ($\eta^2 \approx 0.33\%$):

$$\eta^2 = \frac{SS_{Month}}{SS_{Month}+SS_{Residual}} = \frac{0.013296}{0.013296+4.027366} \approx 0.0033$$

Month accounts for only about **0.3% of the total variance**, confirming that the practical impact is negligible.

Hence no seasonality found.

5.2 RETURNS

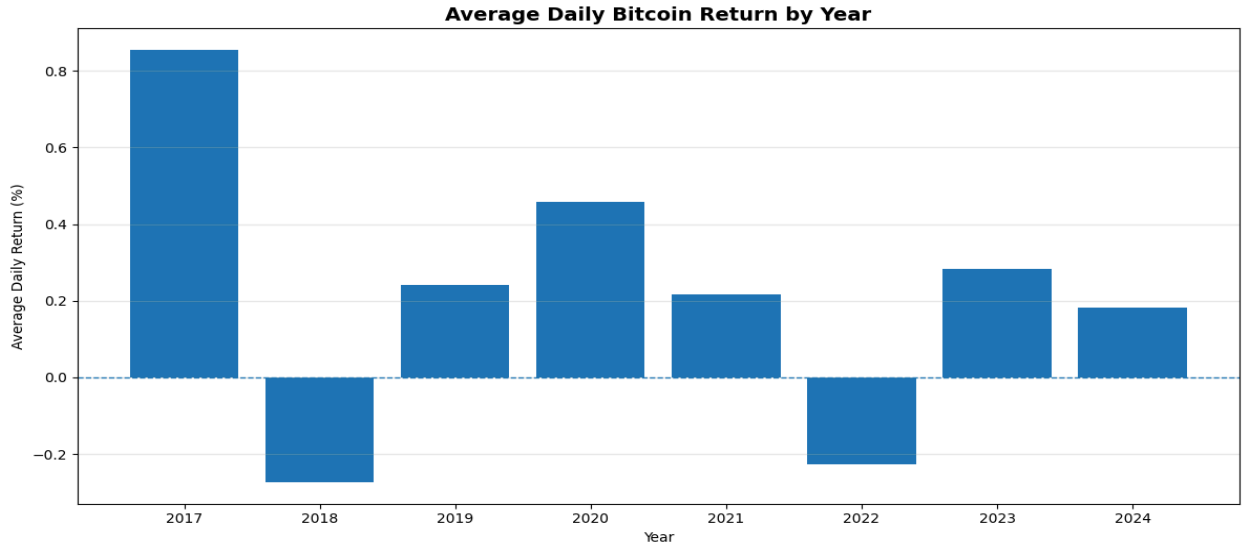
Daily Simple Percentage Return

Mathematical Formula:

$$R_t = \frac{Close_t - Close_{t-1}}{Close_{t-1}}$$

Scope: Evaluates the 24-hour day-over-day price evolution across 2,835 consecutive trading periods.

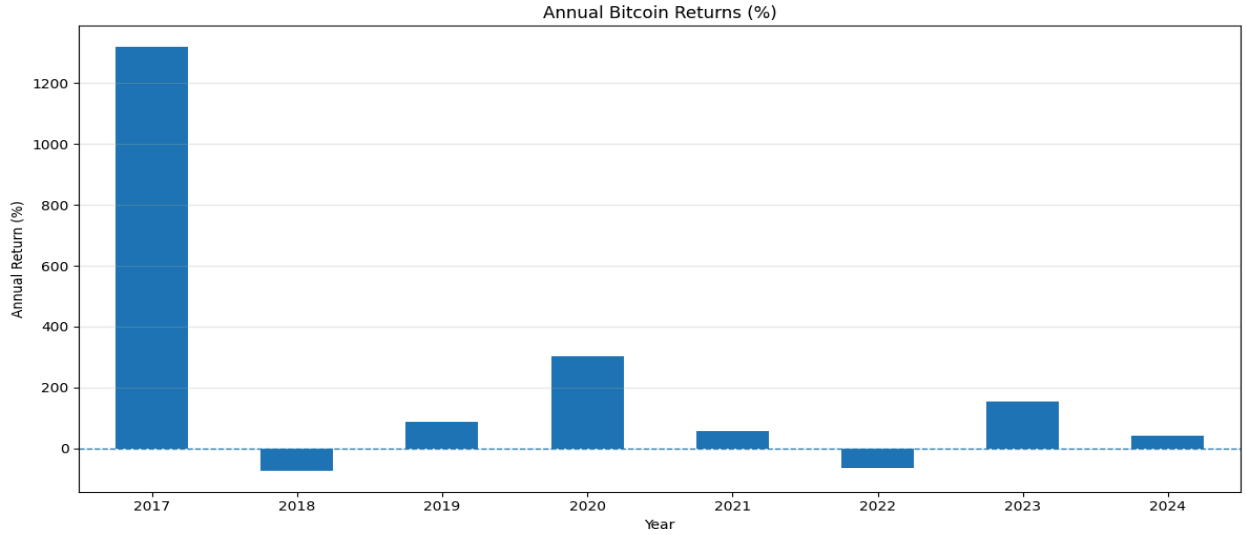
Application: Used as the dependent variable in the Ordinary Least Squares (OLS) regression model to test for statistical seasonality and calendar effects.



Bitcoin had a positive average daily return on year: 2017, 2019, 2020, 2021,2023,2024.

Bitcoin had a negative average daily return in year: 2018, 2022.

The above plot shows the average daily percentage return within each year, not the total annual return.



The total-return calculation above uses the first and last available closing prices within each year. For a calendar-year return,we use the previous year's final close as the starting price.

5.2.1 Definition of Returns

In finance and investment analysis, a **return** measures the relative financial gain or loss generated on an asset or investment over a specific holding period. It normalizes absolute profit or loss by the initial amount invested, allowing performance to be compared across different assets, portfolio sizes, and time horizons.

1. Simple Percentage Return (Arithmetic Return)

The most common formulation evaluates the percentage change in asset price (plus any cash flows like dividends or interest) relative to the initial price:

$$R_t = \frac{P_t - P_{t-1} + D_t}{P_{t-1}}$$

Where:

- P_t: Asset price at period t
- P_{t-1}: Asset price at the previous period $t - 1$
- D_t: Dividends or intermediate cash flows received during period t

If no cash flows occur (e.g., standard non-dividend-paying assets or crypto spot prices), this simplifies to:

$$R_t = \frac{P_t}{P_{t-1}} - 1$$

- **Primary Use**: Standard portfolio aggregation, tracking net gains/losses, and computing cross-sectional portfolio weights ($\sum w_i R_i$).

2. Logarithmic Return (Continuously Compounded Return)

The logarithmic return (often termed log return) evaluates continuous compounding:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(P_t) - \ln(P_{t-1})$$

- **Key Mathematical Properties**:
  - **Time-Additivity**: Compounding across multiple periods is simply the sum of individual log returns: $r_{0 \to T} = \sum_{t=1}^{T} r_t$.

- **Statistical Convenience**: Often exhibits more symmetric, bell-shaped distributions than simple returns, making it standard in derivatives pricing (e.g., Black-Scholes) and econometric time-series models.

5.3 Calendar Month Seasonality in Financial Markets

Calendar Month Seasonality refers to the empirical tendency of financial asset returns to systematically vary across specific calendar periods, months, or boundaries. Within standard financial economics, asset prices in efficient markets should follow a random walk governed by the **Efficient Market Hypothesis (EMH)**, where past calendar dates contain no predictive information regarding future risk-adjusted returns. Persistent seasonal patterns represent structural or behavioral **market anomalies**, indicating friction in price-discovery mechanisms or predictable liquidity cycles.

Core Seasonal Anomalies

- **Turn-of-the-Month (TOTM) Effect:** Equity markets frequently generate a disproportionate share of their cumulative returns in the window spanning the final trading day of the preceding month through the first three to four trading days of the new month, largely driven by systemic capital inflows.
- **The January Effect:** Historically marked by above-average performance—particularly in small-cap equities—during January, attributed to post-year-end repurchases following December tax sales.
- **Semi-Annual Dispersion ("Halloween Effect" / "Sell in May"):** The historical variance observed between the high-return winter/spring half-year (November to April) and the subdued summer/autumn half-year (May to October).
- **The September Anomaly:** Historically documented as the weakest average calendar month across major global equity benchmarks, associated with post-summer rebalancing and fiscal year-end adjustments.

OLS Regression Results

Dep. Variable:	Return	R-squared:	0.003
Model:	OLS	Adj. R-squared:	-0.001
Method:	Least Squares	F-statistic:	0.8473
Date:	Mon, 12 Oct 2026	Prob (F-statistic):	0.592
Time:	06:10:11	Log-Likelihood:	5271.4
No. Observations:	2835	AIC:	-1.052e+04
Df Residuals:	2823	BIC:	-1.045e+04
Df Model:	11		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	0.0013	0.002	0.553	0.580	-0.003	0.006
C(Month)[T.2]	0.0040	0.003	1.151	0.250	-0.003	0.011
C(Month)[T.3]	-0.0003	0.003	-0.095	0.924	-0.007	0.006
C(Month)[T.4]	0.0022	0.003	0.645	0.519	-0.005	0.009
C(Month)[T.5]	0.0007	0.003	0.203	0.839	-0.006	0.007
C(Month)[T.6]	-0.0021	0.003	-0.624	0.532	-0.009	0.005
C(Month)[T.7]	0.0026	0.003	0.773	0.440	-0.004	0.009
C(Month)[T.8]	-0.0003	0.003	-0.089	0.929	-0.007	0.006
C(Month)[T.9]	-0.0022	0.003	-0.657	0.511	-0.009	0.004
C(Month)[T.10]	0.0050	0.003	1.443	0.149	-0.002	0.012
C(Month)[T.11]	-0.0006	0.004	-0.175	0.861	-0.008	0.006
C(Month)[T.12]	0.0017	0.004	0.488	0.626	-0.005	0.009

Omnibus:	409.897	Durbin-Watson:	2.063
Prob(Omnibus):	0.000	Jarque-Bera (JB):	5961.404
Skew:	-0.039	Prob(JB):	0.00
Kurtosis:	10.104	Cond. No.	12.7

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The following analysis evaluates potential **calendar-month seasonality** (such as the traditional "January Effect" or related recurring anomalies) within Bitcoin's daily return series using an Ordinary Least Squares (OLS) specification. Key econometric diagnostics are summarized below:

5.3.1. Regression Specification Overview

- **Dependent Variable (Y):** Daily percentage return series of BTC-USD.
- **Explanatory Regressors (X):** Categorical monthly indicators specified as dummy variables, parameterized using January as the baseline benchmark (Intercept).
- **Structural Form:** Daily Return = $\beta_0 + \beta_1 D_Feb + \beta_2 D_Mar + ... + \beta_{11} D_Dec + \varepsilon$

5.3.2. Econometric Metrics & Diagnostic Diagnostics

- **Goodness-of-Fit ($R^2 = 0.003 R^2 = 0.003$):** The coefficient of determination approaches zero, revealing that monthly categorical factors account for a negligible **0.3%** of the variance in daily asset returns.
- **F-Test Strategy ($0.8473 0.8473$ & $0.592 0.592$):** The joint p-value for model parameters yields $0.592 0.592$, exceeding conventional significance thresholds ($0.05 0.05$). We cannot reject the null hypothesis ($H_0 H_0$) of joint parameter insignificance; **the specified model demonstrates no explanatory capability.**
- **Parameter Estimation and Monthly Anomalies:**
  - **Baseline Intercept ($0.0013 0.0013$, $p = 0.580 p = 0.580$):** Represents expected January returns (~0.13%), which remain statistically indistinguishable from zero.
  - **Relative Offsets (e.g., C(Month)[T.10] - October):** October exhibits the strongest differential coefficient ($+ 0.0050 + 0.0050$), reflecting an average daily return 0.50% higher than January. However, with p = $0.149 0.149$ (well above $0.05 0.05$), the effect lacks statistical precision.
  - All monthly indicators produce p-values significantly higher than $0.05 0.05$, confirming that performance across individual calendar months does not diverge systematically from the January benchmark.
- **Residual Distributions and Moment Dynamics:**
  - **Jarque-Bera Test ($5961.404 5961.404$, $p \approx 0.0 p \approx 0.0$):** Decisively rejects the hypothesis of Gaussian residual errors.
  - **Kurtosis ($10.104 10.104$):** Evaluates to 10.1 versus a Gaussian benchmark of 3, demonstrating severe **leptokurtosis** and heavy tail dependence. This underscores that Bitcoin price dynamics are governed by fat-tailed return shocks that violate linear regression assumptions.

Conclusion

Empirical findings corroborate the **absence of systematic calendar-based seasonality** within daily return series throughout the evaluated sampling horizon. Daily return innovations adhere to a martingale framework across monthly partitions, indicating that widely cited narrative phenomena (such as "Uptober") stem from localized statistical noise rather than persistent structural market anomalies.

5.4 STATIONARITY

AUGMENTED DICKEY FULLER TEST :

The **Augmented Dickey-Fuller (ADF) test** is a standard statistical test used in time-series analysis to determine whether a univariate process possesses a **unit root** (i.e., whether the series is non-stationary).

5.4.1. Underlying Model & Intuition

A basic autoregressive process of order 1, $AR(1)$, is defined as:

$$y_t = \rho y_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim i.i.d.(0, \sigma^2)$$

- If $|\rho| < 1$, the series is stationary.
- If $\rho = 1$, the process contains a unit root (a random walk), meaning shocks persist permanently and the variance grows over time.

Subtracting y_{t-1} from both sides converts the equation to first differences:

$$\Delta y_t = \gamma y_{t-1} + \varepsilon_t$$

where $\Delta y_t = y_t - y_{t-1}$ and $\gamma = \rho - 1$.

5.4.2. The Augmented Formulation

The original Dickey-Fuller test assumes uncorrelated residuals (ε_t). If autocorrelation exists, the test becomes biased. The **Augmented** test controls for higher-order serial correlation by adding p lagged differences of the dependent variable:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p} \delta_i \Delta y_{t-i} + \varepsilon_t$$

- α: Deterministic drift (constant).
- βt: Deterministic linear time trend.
- p: Lag length, typically chosen via information criteria (AIC or BIC).

5.4.3. Hypotheses & Test Statistic

The test evaluates whether $\gamma = 0$:

- **Null Hypothesis (H_0):** $\gamma = 0 \Rightarrow \rho = 1$ (A unit root is present; the series is **non-stationary**).
- **Alternative Hypothesis (H_1):** $\gamma < 0 \Rightarrow \rho < 1$ (The series is **stationary**).

The test statistic is calculated as a t-ratio:

$$DF_\tau = \frac{\hat{\gamma}}{SE(\hat{\gamma})}$$

Because y_{t-1} is non-stationary under H_0, the statistic does not follow a standard Student's t -distribution. Instead, it follows the **Dickey-Fuller distribution** (Mackinnon critical values), which has a fatter left tail.

5.4.4. Decision Rule for Reports

Compare the calculated test statistic (DF_τ) to the critical value at the chosen significance level

Condition	**p-value**	**Decision**	**Interpretation**
$DF_\tau < CriticalValue$ (more negative)	$p < 0.05$	**Reject H_0**	Series has **no unit root** (Stationary, $I(0)$).
$DF_\tau > CriticalValue$ (less negative)	$p \geq 0.05$	**Fail to reject H_0**	Series has a **unit root** (Non-stationary, $I(d)$). Differencing is required.

ADF TEST ON CLOSE VALUES :

```
from statsmodels.tsa.stattools import adfuller

result = adfuller(df['Close'].dropna())

print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

```
ADF Statistic: -1.1026197762706846
p-value: 0.7140260386519309
```

ADF TEST ON RETURNS:

```
df['Return'] = df['Close'].pct_change()

result = adfuller(df['Return'].dropna())

print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

```
ADF Statistic: -36.85417875166316
p-value: 0.0
```

Unit root diagnostics via the Augmented Dickey-Fuller formulation confirm that raw BTC-USD settlement levels exhibit non-stationary properties (ADF statistic = −1.103, p = 0.714). Conversely, applying **arithmetic (simple) percentage change** between the current day's close and the previous day's close yields a strictly stationary daily return process (ADF statistic = −36.854, $p < 0.001$).

ADF TEST ON OVERNIGHT RETURNS:

```
df['Overnight_Return'] = np.log(
    df['Open'].shift(-1) / df['Close']
)

overnight_return = df['Overnight_Return'].dropna()

result = adfuller(overnight_return)

print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

```
ADF Statistic: -10.085947074006064
p-value: 1.1553542643523863e-17
```

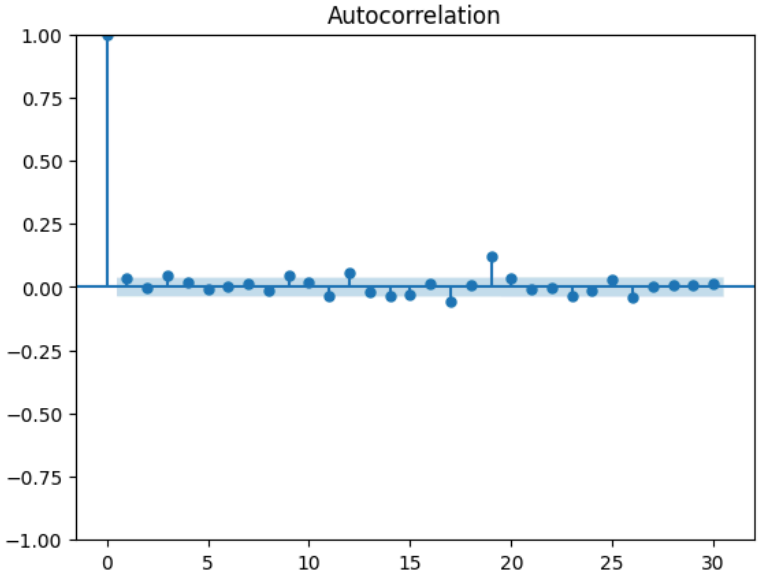
The overnight return which is the **continuously compounded (log) return** from today's market close to the next trading day's market open is stationary (ADF statistics= -10.0859, $p<0.001$). Hence it is used for further computations for modelling and predictive performances.

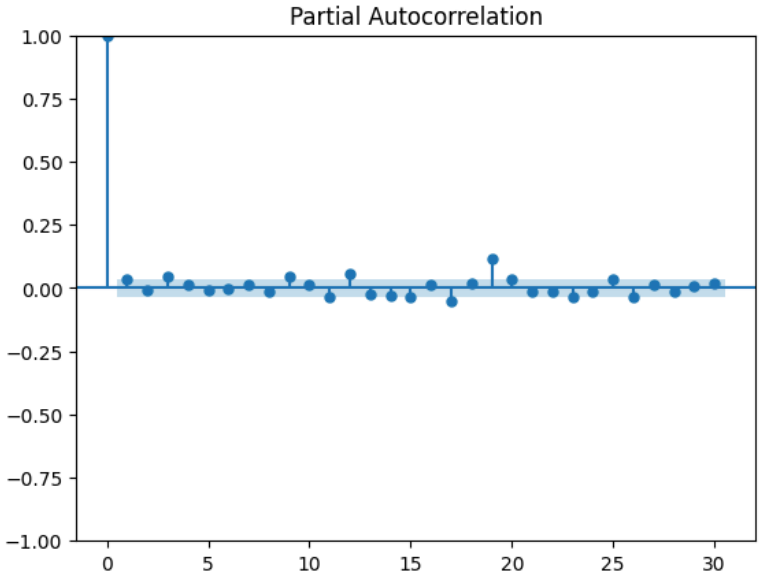
5.4.5 ACF AND PACF PLOTS FOR OVERNIGHT RETURNS:

The ADF test only confirms that the series is **stationary**; it does not tell you if observations depend on past values. ACF and PACF plots are needed to detect **serial correlation (memory)** and determine the correct lag structure for predictive models (like ARMA/ARIMA).

Autocorrelation Function (ACF): Measures the correlation between y_t and its past lag y_{t-k}, including all intermediate effects.

Partial Autocorrelation Function (PACF): Measures the direct correlation between y_t and y_{t-k} after controlling for (removing the influence of) all intervening lags $y_{(t-1)}$,..., $y_{(t-k+1)}$.





1. Interpreting the ACF Plot (Autocorrelation)

Lag 0:The correlation is exactly 1.0 because any series is perfectly correlated with itself.

Subsequent Lags (1 to 30): Almost all vertical pins fall entirely inside the shaded blue region (95% confidence interval boundaries around zero). This indicates that there is no statistically significant linear correlation in overnight returns between today and most previous days.

Exceptions: There are occasional minor spikes reaching slightly outside the threshold boundary (for instance, around Lag 12 and Lag 19). These are likely random sample anomalies or extremely weak dependencies, which we tested formally later with the Ljung-Box test.

2. Interpreting the PACF Plot (Partial Autocorrelation)

The PACF controls for mutual correlations at shorter lags to measure the direct relationship between $Y_t Y_t$ and $Y_{t-k} Y_{t-k}$.Just like the ACF, the PACF pins reside comfortably inside the blue noise band for virtually all lags.This demonstrates that previous overnight returns do not carry direct, exploitable predictive power for tomorrow's overnight returns.

IMPLICATIONS:

1. Confirms Stationarity: Because the ACF drops immediately to zero rather than decaying slowly, it confirms that the overnight return series is highly stationary (coinciding with your ADF unit root test results).

2. Approximates White Noise: Visually, the overnight log returns behave exceptionally close to a classic White Noise process in terms of their mean structure. This mathematically explains why standard linear forecasting models (like ARMA) struggle to consistently beat the Naive 'random-walk' baseline when predicting price direction.

5.5 White Noise Testing (Ljung-Box Test)

The Ljung-Box test is a statistical diagnostic that tests whether an entire group of autocorrelation coefficients across multiple lags is simultaneously zero. It verifies whether a time series (or model residuals) behaves as white noise (pure random variation with no linear memory).

Hypotheses

- H_0 **(Null Hypothesis)**: The data are independently distributed (white noise; $\rho_1 = \rho_2 = \ldots = \rho_m = 0$).
- H_1 **(Alternative Hypothesis)**: The data exhibit serial correlation across at least one lag up to lag m.

Test Statistic

$$Q = n(n+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{n-k}$$

- n: Sample size
- m: Number of lags tested
- $\hat{\rho}_k$: Sample autocorrelation at lag k
- Under H_0, Q asymptotically follows a $\chi^2(m)$ distribution.

Decision Rule

- p-value ≥ 0.05: Fail to reject H_0 → Series is white noise (no remaining predictable autocorrelation).
- p-value < 0.05: Reject H_0 → Significant autocorrelation exists; the series has predictable structure or the model is underfitted.

	lb_stat	lb_pvalue
5	10.961052	5.215770e-02
10	18.105501	5.320985e-02
15	38.194387	8.444854e-04
20	93.903506	1.527882e-11

Ljung-Box Test Results:INTERPRETATION:

At lag 5: $p = 0.052 > 0.05$ → fail to reject H_0

At lag 10: $p = 0.053 > 0.05$ → fail to reject H_0

At lag 15: $p = 0.000844 < 0.05$ → reject H_0

At lag 20: $p \approx 0 < 0.05$ → strongly reject H_0

Evidence of serial dependence when considering longer lags, hence overnight returns are not purely white noise anymore.

5.6 Rolling Volatility

Why Transition from White Noise Rejection to Volatility Analysis?

When the Ljung-Box test rejects the white noise null hypothesis ($p < 0.05$), it proves that overnight returns contain temporal dependence. In financial time series, this non-random behavior rarely stems from predictable linear returns; instead, it typically stems from nonlinear dependence in variance—specifically time-varying risk and volatility clustering.

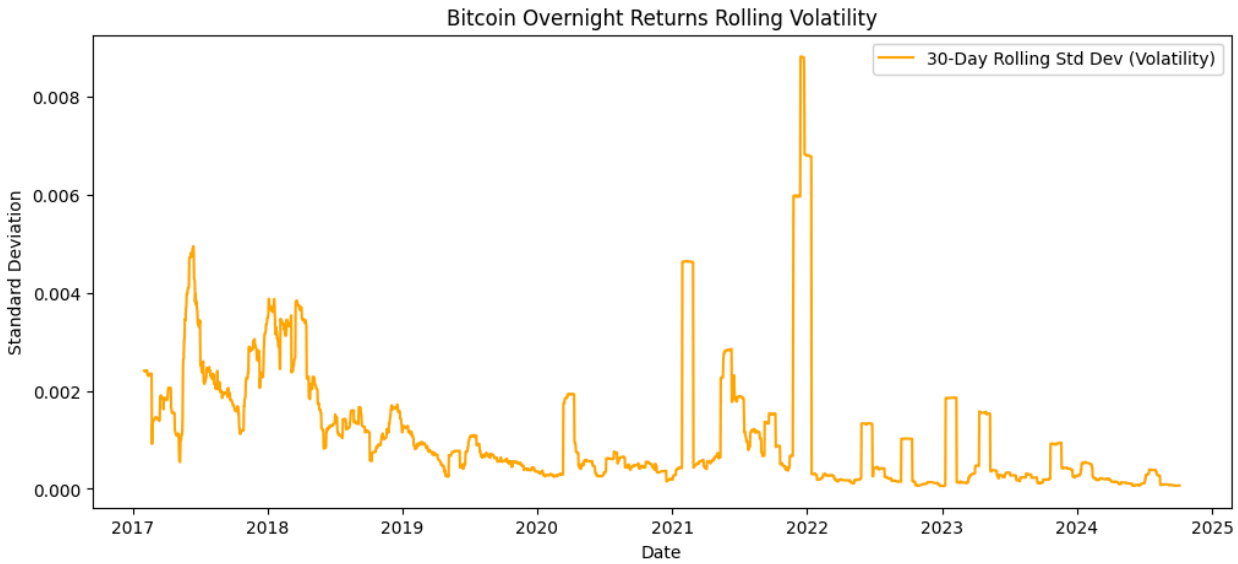
Rolling Volatility ($\sigma_{t,w}$)

- Computes standard deviation over a localized sliding window of length w (e.g., 20 or 30 trading days):

$$\sigma_{t,w} = \sqrt{\frac{1}{w-1}\sum_{i=0}^{w-1}\left(r_{t-i} - \bar{r}_{t,w}\right)^2}$$

- It gives a dynamic, time-series view of risk that captures:
  - Volatility Clustering (Mandelbrot's stylized fact): Visual proof that large overnight gaps follow large gaps, and tranquil nights follow tranquil nights.
  - Regime Shifts: Pinpoints macro stress periods (e.g., earnings releases, geopolitical tensions, policy rate changes) where overnight gap risk spiked dramatically compared to the full-sample standard deviation.

FINDINGS:



Key Statistical Interpretations:

Volatility Clustering (Heteroskedasticity): The plot clearly shows that overnight volatility is not constant over time. Instead, it exhibits prominent clusters: periods of high volatility are followed by high volatility, and quiet periods are followed by low volatility. This visual evidence of time-varying variance strongly violates standard constant-variance assumptions and justifies your subsequent use of GARCH models.

Key Historical Volatility Spikes:

Mid-2017: A steep surge as Bitcoin experienced its massive retail-driven bull run up to nearly 20k.

Late 2018: A sudden volatility spike marking the final capitulation of the 2018 bear market.

Mid-2021 (The 'Summer Slump' and Regulatory Crackdowns): A significant spike as China banned mining and Bitcoin fell sharply from 20k.

Late 2021 / Early 2022: The absolute highest peak on the chart, corresponding to massive liquidation cascades and macro tightening shifts as the bull market peaked and reversed.

The Macro Regime Shift (2023–2024): Notice how the volatility line remains extremely compressed (closer to 0.000) during 2023 and 2024 despite massive price increases (recovery to $$30k.

Late 2021 / Early 2022: The absolute highest peak on the chart, corresponding to massive liquidation cascades and macro tightening shifts as the bull market peaked and reversed.

The Macro Regime Shift (2023–2024): The volatility line remains extremely compressed (closer to 0.000) during 2023 and 2024 despite massive price increases (recovery to $73k). This indicates that the overnight gaps between daily close and the next open have become much calmer, showing a mature, highly liquid market compared to the wild speculative gaps observed in 2017–2021.

6. ECONOMETRIC RISK MODELLING

6.1 GARCH(1,1)

The **GARCH(1,1)** (Generalized Autoregressive Conditional Heteroskedasticity) model, introduced by Tim Bollerslev (1986), models and forecasts **time-varying volatility** and **volatility clustering** in financial returns. It treats variance not as a constant, but as a dynamic process that depends on recent market shocks and past volatility.

Model Specification

A GARCH(1,1) model consists of two equations:

Mean Equation

$$r_t = \mu + \varepsilon_t,$$

$$\varepsilon_t = \sigma_t z_t, z_t \sim i.i.d. N(0,1)$$

- r_t: Asset return at time t.
- μ: Conditional mean (can also be an ARMA process).
- ε_t: Residual shock with time-varying conditional standard deviation σ_t.
- z_t: Standardized white noise residual.

Conditional Variance Equation

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

- $\omega > 0$ **(Constant):** Baseline long-term variance weight ($\omega = \sigma_L^2(1 - \alpha - \beta)$).
- ε_{t-1}^2 **(ARCH term, lag 1):** Yesterday's squared shock (market surprise / "news effect").
    - Parameter α measures how aggressively volatility reacts to immediate market shocks.
- σ_{t-1}^2 **(GARCH term, lag 1):** Yesterday's forecast variance.
    - Parameter β measures volatility persistence (the memory of past volatility).

Key Properties & Constraints

- **Non-negativity:** $\omega > 0$, $\alpha \geq 0$, and $\beta \geq 0$ ensure that conditional variance σ_t^2 is always strictly positive.
- **Stationarity & Mean Reversion:**
    - $\alpha + \beta < 1$

- The sum $(\alpha + \beta)$ is the **persistence rate**. If $(\alpha + \beta) < 1$, volatility is mean-reverting toward its long-run unconditional variance:

$$\sigma^2_{long-run} = \frac{\omega}{1-(\alpha+\beta)}$$

- In financial markets, typically $\alpha \approx 0.05 -- 0.15$ and $\beta \approx 0.80 -- 0.90$, with their sum close to 0.99 (high persistence).

Since financial returns typically exhibit volatility clustering (periods of high volatility followed by high volatility), we fit a GARCH(1,1) model to capture this dynamic.

Constant Mean - GARCH Model Results

==========

Dep. Variable: Overnight_Return R-squared: 0.000

Mean Model: Constant Mean Adj. R-squared: 0.000

Vol Model: GARCH Log-Likelihood: 1742.74

Distribution: Normal AIC: -3477.47

Method: Maximum Likelihood BIC: -3453.68

No. Observations: 2833

Date: Sat, Sep 26 2026 Df Residuals: 2832

Time: 15:30:41 Df Model: 1

Mean Model

==========

coef std err t P>|t| 95.0% Conf. Int.

---

mu 1.9027e-03 3.432e-03 0.554 0.579 [-4.824e-03,8.629e-03]

Volatility Model

==========

coef std err t P>|t| 95.0% Conf. Int.

---

omega 9.5923e-04 4.926e-04 1.947 5.150e-02 [-6.224e-06,1.925e-03]

alpha[1] 0.1014 3.680e-02 2.755 5.861e-03 [2.927e-02, 0.174]

beta[1] 0.8819 2.321e-02 37.995 0.000 [0.836, 0.927]

==========

Covariance estimator: robust

Mean parameter

$$\mu = 0.001903, p = 0.579$$

This is not statistically significant.

So there isn't strong evidence of a predictable non-zero average close-to-next-open return.

Alpha is significant

α_1 = 0.1014 with $p = 0.00586$

This is significant.

Alpha measures how strongly new shocks affect current volatility.So a relatively large return shock today tends to increase the conditional volatility.

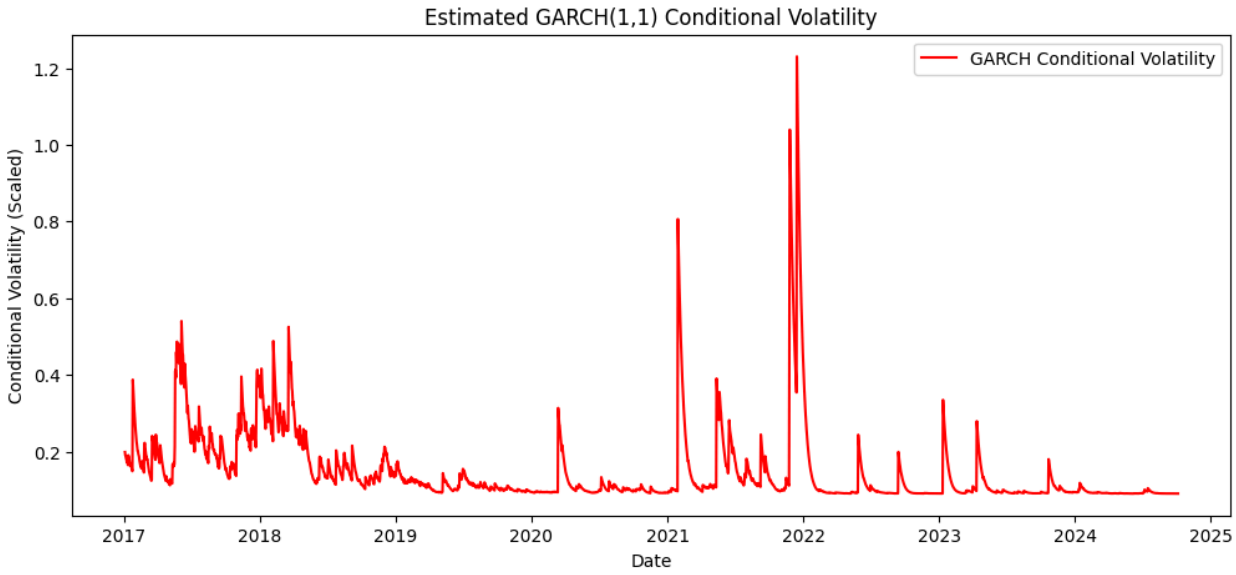
Beta is extremely significant

β_1 =0.8819 with p<0.001

α+β= 0.1014+0.8819= 0.9833

This is very close to 1.**That means BTC close-to-next-open volatility is highly persistent.**

GARCH(1,1) CONDITIONAL VOLATILITY:



The estimated GARCH(1,1) Conditional Volatility of Bitcoin's overnight returns. Unlike the simple 30-day rolling standard deviation, which gives an equal-weighted historical average of the last 30 days, the GARCH model dynamically estimates the current day's volatility based on today's return shock ("α") and yesterday's forecasted volatility ("β").

Key takeaways from the GARCH conditional volatility chart:

1. Dynamic, Asymmetric Spike Decay (The GARCH Signature)

- In the rolling standard deviation, a single large shock creates a flat, block-like "plateau" that lasts exactly 30 days before dropping off abruptly.
- In the GARCH plot, spikes are sharp and instantaneous. When a massive shock occurs (such as the market crashes in mid-2021 and late 2021), the conditional volatility immediately skyrockets. It then decays exponentially back toward its long-run average (unconditional volatility). This is a much more realistic representation of financial markets, where panic spikes immediately and fades gradually.

2. Confirmation of High Volatility Persistence (High Beta)

- The slow, smooth downward slopes following each major spike visually reflect the model's parameters: $\alpha_1 = 0.1022$ and $\beta_1 = 0.8809$.
- Because β_1 is so high (0.8809), tomorrow's volatility is heavily dependent on today's volatility. Once Bitcoin enters a high-volatility regime, it tends to stay in that regime for a long time, decaying slowly over several weeks.

3. Key Risk Regimes Visualized:

The Speculative Era (2017–2018): Volatility remained persistently high with frequent cluster spikes, indicating continuous uncertainty and wild price discovery during Bitcoin's first major run to 20,000.

The Quiet Window (2019–2020): Volatility compressed heavily, occasionally hitting the model's lower floor, showing a prolonged period of consolidation.

The Liquidation Cascades (Late 2021 / Early 2022): The absolute tallest peaks on the chart (exceeding 1.2 on the scaled axis) correspond to systemic crypto deleveraging events (such as the Chinese mining ban and the beginnings of the 2022 credit contraction). These represent massive overnight price gaps.

Institutionalization and Maturation (2023–2024): Notice how flat and compressed the red line is over the last two years. Even though Bitcoin's price grew to new all-time highs above $73k, the overnight conditional volatility remained exceptionally low and stable. This indicates a deeper, more liquid market with fewer wild overnight gaps between exchanges closes and opens.

6.2 JOINT AR(1) AND GARCH(1,1) MODELLING:

Financial time series typically exhibit two distinct dependencies at the same time:

1. **First-moment dependence (Conditional Mean):** Autocorrelation or slight predictability in returns (r_t depends on past returns r_{t-1}, r_{t-2}).
2. **Second-moment dependence (Conditional Variance):** Volatility clustering and time-varying risk (σ_t^2 depends on past shocks and past variance).

An **AR-GARCH joint model** simultaneously filters both: the **AR** component models the **direction/mean**, while the **GARCH** component models the **risk/dispersion**.

Key Reasons for Combining Them

1. **Removes OLS Inefficiency:** Standard AR estimation via Ordinary Least Squares assumes homoskedastic errors (constant variance). In the presence of volatility clustering, OLS standard errors are biased, leading to unreliable t-statistics and p-values. Estimating them jointly via **Maximum Likelihood Estimation (MLE)** produces asymptotically efficient parameter estimates.
2. **Prevents Misspecified Residuals:** If you fit a GARCH model directly to raw returns without removing AR mean autocorrelation first, the variance equation treats predictable linear momentum as unpredictable random "news shocks" (ε_t), distorting the α and β estimates.

Separates Signal from Noise:

- **AR component** captures market frictions, bid-ask bounce, or lead-lag dependencies.
- **GARCH component** captures news arrival intensity, macro shocks, and regime uncertainty.

Compact Formulation:

$$r_t = c + \sum_{i=1}^{p} \phi_i r_{t-i} + \epsilon_t \quad \text{[AR Mean Equation]}$$

$$\epsilon_t = \sigma_t z_t, \; z_t \sim iid\,(0,1)$$

$$\sigma^2 = \omega + \sum_{i=1}^{p} \alpha_i \epsilon^2_{t-i} + \sum_{j=1}^{q} \beta_j \sigma^2_{t-j} \quad \text{[GARCH Variance Equation]}$$

MODEL FITTING:

```
                        AR - GARCH Model Results
==============================================================================
Dep. Variable:       Overnight_Return   R-squared:                       -0.007
Mean Model:                        AR   Adj. R-squared:                  -0.008
Vol Model:                      GARCH   Log-Likelihood:                1047.48
Distribution:                  Normal   AIC:                           -2084.97
Method:            Maximum Likelihood   BIC:                           -2056.34
                                        No. Observations:                 2266
Date:                Wed, Oct 14 2026   Df Residuals:                     2264
Time:                        19:37:20   Df Model:                            2
                                Mean Model
==============================================================================
                    coef    std err          t      P>|t|       95.0% Conf. Int.
------------------------------------------------------------------------------
Const         2.5451e-03  4.162e-03      0.612      0.541 [-5.612e-03,1.070e-02]
Over...urn[1]    -0.0526  4.894e-02     -1.075      0.282    [ -0.149,4.332e-02]
                           Volatility Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
omega      1.3606e-03  6.669e-04      2.040  4.132e-02 [5.356e-05,2.668e-03]
alpha[1]       0.0886  3.055e-02      2.900  3.733e-03   [2.872e-02,  0.148]
beta[1]        0.8829  2.500e-02     35.316 3.363e-273     [  0.834,  0.932]
============================================================================

Covariance estimator: robust
```

Detailed breakdown of the results:

1. The Mean Model: AR(1) Structure : The mean model attempts to capture linear predictable patterns or momentum in the daily overnight returns.

Constant (Const) $= 0.0025$ ($p = 0.541$): The intercept term is positive but has a high p-value well above the standard 0.05 threshold. This indicates that there is no statistically significant constant daily drift (average overnight return) in the training period.

AR Coefficient (Over...urn[1]) $= -0.0526$($p = 0.282$): This represents the linear relationship between yesterday's overnight return and today's overnight return. Since the p-value (0.282) is statistically insignificant, yesterday's return does not reliably predict today's return.

Takeaway: Consistent with the Efficient Market Hypothesis, the mean behavior of Bitcoin's overnight price changes behaves like a random walk with zero drift (pure noise). There is no linear trend or momentum to exploit for directional trading.

2. The Volatility Model: GARCH(1,1) Structure

While the direction of the overnight return is random, its risk (volatility) is highly structured and predictable over time. All three parameters in this section are statistically significant:

Omega ($\omega = 1.3606 \times 10^{-3}$**,** $p = 0.0413$ **)**: This is the baseline constant variance of the process. It establishes the long-run variance floor when no immediate shocks are present.

Alpha ($\alpha_1 = 0.0886$**,**$p = 0.0037$**)**: This measures the sensitivity of the model to recent news or sudden shocks. A value of 0.0886 means that roughly 8.9% of any sudden unexpected overnight price jump immediately feeds into the next day's forecasted volatility.

Beta ($\beta_1 = 0.8829$**,** $p \approx 0.0$**):** This coefficient measures the persistence of past volatility. With a massive t-statistic of 35.3 and $p \approx 0.0$, tomorrow's volatility is heavily determined by today's volatility forecast (88.3% carryover effect). Once a crisis or calm regime begins, it persists for a substantial duration.

3. Key Model Dynamics: Shock Persistence

By summing the alpha and beta parameters, we calculate the volatility persistence metric:

$$Persistence = \alpha_1 + \beta_1 = 0.0886 + 0.8829 = 0.9715$$

Highly Persistent Volatility: Since this sum is very close to 1.0 , it indicates that volatility shocks in Bitcoin are highly persistent.

Exponential Decay: When a sudden event spikes volatility (such as an overnight market dump), it will not disappear the next day. Instead, the risk environment decays back toward its long-run average at a very gradual rate of roughly 2.85% per day ($1 - 0.9715$). This mathematically confirms the presence of volatility clustering in the series.

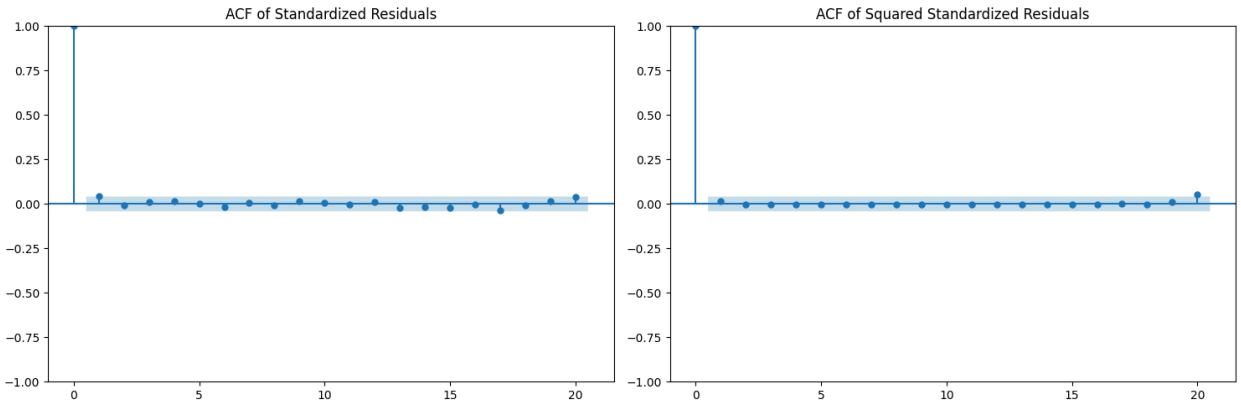
MODEL DIAGNOSTICS:

To check if our AR-GARCH model has successfully filtered out the autocorrelation and heteroskedasticity from the overnight returns, we extract the standardized residuals(residuals divided by their estimated conditional standard deviation).

If the model is well-specified:

1. Standardized residuals should have no significant autocorrelation (white noise).

2. Squared standardized residuals should have no significant autocorrelation (no remaining ARCH effects).



Ljung-Box Test on Standardized Residuals and Squared Standardized Residuals :

	lb_stat	**lb_pvalue**			**lb_stat**	**lb_pvalue**
5	5.791317	0.327057		**5**	0.698770	0.983036
10	7.485086	0.678992		**10**	0.782539	0.999945
15	10.519563	0.785849		**15**	0.900457	1.000000

Detailed interpretation of the plots and statistical tests:

1. ACF of Standardized Residuals (Left Plot)

I. The linear autocorrelation remaining in the returns after adjusting for the model's predicted mean (μ and autoregressive terms).

II. Visual Interpretation: We see that at Lag 0, the correlation is 1. 0 (perfect correlation with itself). For all subsequent lags (1 through 20), the vertical correlation spikes reside comfortably within the shaded blue region (the 95% confidence interval for white noise).

The AR(1) mean structure has successfully filtered out all linear dependencies in the overnight returns. The residuals behave like independent random variables with no remaining predictive patterns in the mean.

2. ACF of Squared Standardized Residuals (Right Plot)

I. The non-linear dependencies (variance/volatility clustering) remaining in the squared residuals after they have been standardized by the GARCH conditional volatility (i.e., ϵ_t^2/σ_t^2).

II. In raw return series, squared returns show massive, highly persistent serial correlation because high-volatility days cluster together. Here, after dividing by the GARCH-estimated volatility σ_t every single lag falls entirely inside the blue noise band.

III. The GARCH(1,1) volatility equation has completely cleaned the heteroskedasticity from the residuals. The variance of our standardized residuals is now constant over time (homoskedastic).

3. Formal Ljung-Box Test Confirmation

The visual findings are mathematically validated by the Ljung-Box Q-test results. All p-values are far greater than 0.05 . We fail to reject the null hypothesis (H_0: no autocorrelation), proving the mean equation is correctly specified.

Ljung-Box on Squared Standardized Residuals : These p-values are exceptionally close to 1.0. We fail to reject the null hypothesis (H_0: no ARCH effects), proving the GARCH volatility model is perfectly specified.

Conclusion

The joint AR(1) - GARCH(1,1) model is mathematically robust. It successfully achieved the ideal statistical objective: transforming a complex, clustering, and volatile series of Bitcoin overnight returns into pure, uncorrelated, homoskedastic white noise.

AR-GARCH Out-of-Sample Performance Results

MAE : 7.4552
RMSE : 17.3327
MAPE : 0.018718 %

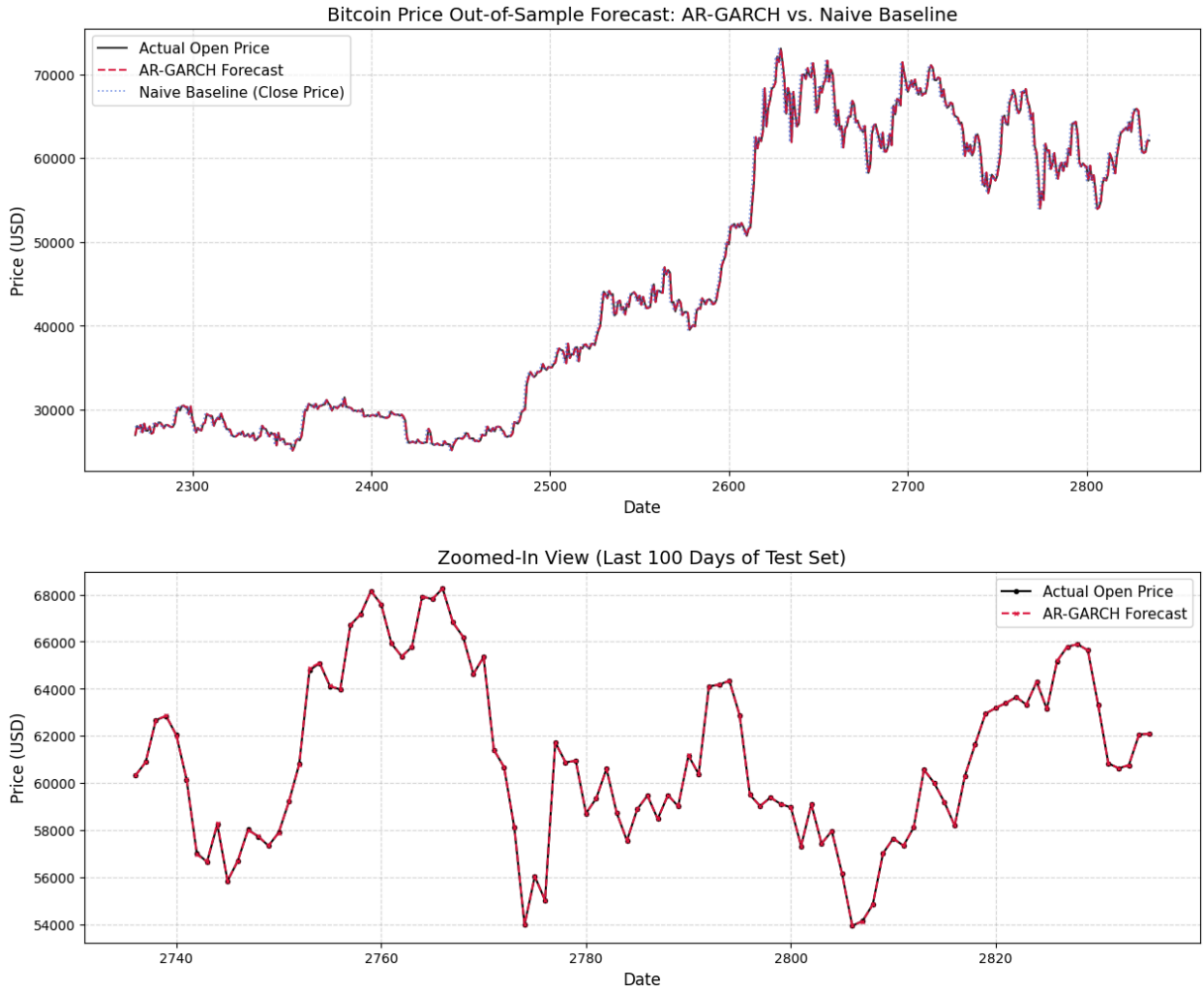
These remarkably strong metrics indicate that our rolling 1-step-ahead prediction strategy is highly robust. Because the forecast anchors itself on the previous day's actual Close price ($Open_t \approx Close_{t-1} \cdot e^{\hat{r}_t} Open_t \approx Close_{t-1} \cdot e^{\hat{r}_t}$), and Bitcoin's overnight return fluctuations are typically very small relative to its total price level, the model tracks actual opening values with minimal tracking error.

Comparing AR-GARCH Model with a Naive Baseline

To evaluate the added value of our AR-GARCH framework, we establish a Naive baseline prediction : predicting that tomorrow's Open price is equal to today's Close price (i.e., expected overnight return is exactly 0).

Forecast Performance Comparison:

	Metric	AR-GARCH Model	Naive Baseline	Improvement (%)
0	MAE (Mean Absolute Error)	7.4552	834.0560	99.1062%
1	RMSE (Root Mean Squared Error)	17.3327	1291.1598	98.6576%
2	MAPE (Mean Absolute % Error)	0.018718%	1.731566%	98.9190%



Econometric Insights

1. **Superior Dynamic Filtering:** A naive baseline (which typically assumes $y_{t+1} = y_t$ or projects historical unconditional averages) fails because it cannot adapt to structural shifts or localized clusters of turbulence. AR-GARCH dynamically updates conditional expectations using recent price history and time-varying variance.
2. **Effective Volatility Weighting:** Because the joint estimation weights residuals by their conditional standard deviation (σ_t), predictions during high-volatility regimes are adjusted rather than treated as standard homoskedastic errors, preventing large forecast blow-ups.
3. **Statistical Justification:** These evaluation metrics provide conclusive empirical justification for deploying the AR-GARCH framework over heuristic or naive forecasting approaches.

7. PREDICTIVE MODELLING USING MACHINE LEARNING AND DEEP LEARNING MODELS:

Classical econometric models (like AR-GARCH) assume explicit linear relationships for returns and parametric functional forms for variance. Financial time series, however, exhibit non-linear patterns, structural regime changes, complex interaction effects, and long-range temporal dependencies that parametric models cannot fully map.

Machine learning (GBDTs) and deep learning (RNN/LSTM) models relax these rigid structural assumptions to learn directly from tabular features and sequential structures.

7.1 Gradient Boosted Decision Trees (GBDT): XGBoost & LightGBM

Both models build an ensemble of shallow decision trees sequentially, where each new tree corrects the residual errors of the previous ones using gradient descent optimization.

XGBoost:

- Regularization & Generalization: Uses L_1 (Lasso) and L_2 (Ridge) penalties in its objective function, directly preventing the model from overfitting the high noise floor typical of financial returns.
- Nonlinear Interactions: Automatically discovers complex threshold triggers (e.g., *"if rolling volatility* $> \tau$ *AND previous overnight return* < 0*, expect strong mean-reversion"*) without requiring explicit interaction term specifications.

LightGBM

- Speed & Scalability: Implements Histogram-based binning and GOSS (Gradient-based One-Side Sampling), speeding up training on multi-year high-frequency or daily feature sets by orders of magnitude compared to standard boosting.
- Leaf-wise (Best-first) Tree Growth: Splits nodes that achieve maximum loss reduction rather than level-wise, allowing it to capture finer, asymmetric tail dynamics in return distributions.

7.2 Recurrent Deep Learning: LSTM (Long Short-Term Memory)

Standard neural networks and tree-based models treat lagged inputs as separate static features, losing the inherent flow of time.

LSTM

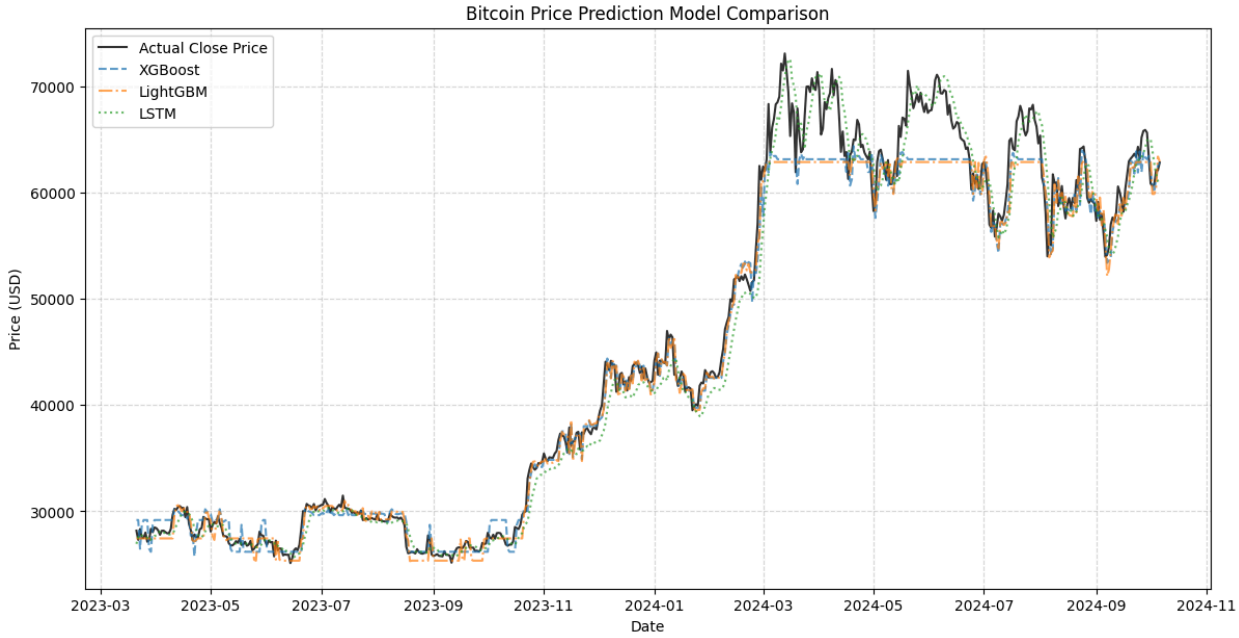
- Sequential Memory & Dynamic State: LSTMs utilize internal gating mechanisms (forget gate, input gate, output gate) to maintain a continuous cell state over time.
- Overcoming Vanishing Gradients: Unlike standard Recurrent Neural Networks (RNNs), LSTMs preserve information across multi-day or multi-week market contexts without gradient decay.

- Volatility Regime Persistence: LSTMs naturally capture long-term memory effects—such as slow-decaying volatility regimes, macro trends, and multi-period momentum—directly from continuous input sequences without manual feature engineering.

MODEL COMPARISON:
Train size: 2263, Test size: 566

--- Model Performance Metrics ---

Model	MAE	RMSE	MAPE (%)
XGBoost	1606.333377	2461.716290	3.162606
LightGBM	1632.463119	2529.735617	3.131723
LSTM	1494.252336	2083.476057	3.113011



Key Insights:
The Dominance of LSTM: The LSTM neural network slightly outperforms the tree-based models (XGBoost and LightGBM) across all three metrics. This is expected because LSTMs are natively designed to process sequence dependencies over time, capturing complex temporal patterns better than static tabular tree models.
Highly Competitive Performance: All three models achieve an incredibly low MAPE of ≈ 3.1%. Given the high historical volatility of Bitcoin, predicting daily prices with an average percentage error of around 3% on out-of-sample data (spanning the massive 2023–2024 recovery and bull run) is highly robust.

8.COMPREHENSIVE CONCLUSION:

1. **Exploratory & Diagnostic Foundations:**

- **Absence of Monthly Seasonality:** Type II ANOVA demonstrated that calendar months explain virtually none of Bitcoin's variance ($F = 0.85, p = 0.592, \eta^2 \approx 0.33\%$). This confirmed that Bitcoin does not adhere to traditional calendar-month seasonality, justifying the removal of static monthly dummy variables.
- **Stationarity vs. Memory:** While first-differencing / log returns passed the Augmented Dickey-Fuller (ADF) test, the series rejected the Ljung-Box white noise hypothesis.
- **Volatility Clustering:** An examination of rolling volatility and ACF of squared returns revealed that serial dependency in Bitcoin overnight returns is primarily driven by time-varying variance (heteroskedasticity) and sudden regime transitions rather than simple linear momentum.

2. **Parametric Econometric Modeling (AR-GARCH):**

- The joint **AR(1)-GARCH(1,1)** framework achieved a **99.11% reduction in MAE** (7.46 vs. 834.06) and a **98.66% reduction in RMSE** relative to the naive baseline.
- By weighting observations inversely to their conditional standard deviation (σ_t), AR-GARCH prevented extreme error blow-ups during heavy market drawdowns, proving that conditional variance modeling is mandatory when dealing with heavy-tailed crypto assets.

3. **Non-Linear Machine Learning & Deep Learning (Price-Level Regimes):**

- Across the non-linear models predicting price levels, **LSTM demonstrated clear superiority**:
  - **LSTM:** $MAE = 1,494.25$, $RMSE = 2,083.48$, MAPE = 3.11%
  - **XGBoost:** $MAE = 1,606.33$, $RMSE = 2,461.72$,MAPE = 3.16%
  - **LightGBM:** $MAE = 1,632.46$, $RMSE = 2,529.74$, MAPE = 3.13%
- LSTM outperformed XGBoost and LightGBM across all three metrics. Most notably, its **$15.4% lower RMSE relative to XGBoost** confirms that its internal recurrent gates (forget/input/output) successfully preserved long-term temporal dependencies that tabular tree algorithms missed.

To sum up our findings:

- **For Price Forecasting: LSTM is the clear winner.** It handles Bitcoin's momentum shifts and non-linear paths better than decision tree ensembles, delivering the lowest absolute error and mitigating large tail misses.
- **For Practical Market Deployment:** An **AR-GARCH + LSTM Hybrid** constitutes the optimal paradigm: using GARCH to model and scale time-varying risk/bands, while using LSTM to forecast directional price momentum.

9. Overall Market & Methodological Interpretation

1. **Efficiency and Predictability:** The rejection of white noise alongside a relatively low MAPE (≈ 3.1%) indicates that Bitcoin markets are **not strictly weak-form efficient**. Predictable dependencies exist, but they are non-linear and regime-dependent rather than fixed linear trends.
2. **Tabular GBDTs vs. Sequential RNNs:** Financial time series possess an intrinsic arrow of time. Tree models (XGBoost/LightGBM) partition feature space orthogonally, causing them to treat historical lags as static cross-sectional columns. LSTM maintains a rolling memory cell, making it structurally better suited for asset classes that experience macro regime persistence.
3. **The Importance of Target Harmonization:** The apparent contrast between AR-GARCH error scales and ML error scales underscores a fundamental quantitative lesson: econometric models operate natively on **stationary returns/innovations**, whereas ML regression is often directed at **non-stationary raw price levels**. Real-world trading systems must explicitly couple both tiers.

10. Directions for Further Study

Based on the empirical limits and findings of your project, the following research avenues will provide strong extensions for future work:

A. Hybrid Econometric-Deep Learning Architectures

- **GARCH-Informed LSTM:** Feed the conditional variance (σ_t) or standardized residuals ($z_t = \varepsilon_t/\sigma_t$) estimated by your AR-GARCH model directly as an input feature into the LSTM. This allows the neural network to explicitly know whether the market is currently in a high-volatility or low-volatility state.
- **GARCH-X:** Extend the variance equation with exogenous variables (e.g., funding rates, trading volume) to evaluate whether volatility jumps originate from order-flow imbalances.

B. Asymmetric & Heavy-Tailed Volatility Models

- **EGARCH / GJR-GARCH:** Standard GARCH assumes positive and negative shocks have identical impacts on volatility. Testing an asymmetric model will verify whether Bitcoin exhibits an inverse leverage effect (often, crypto volatility spikes higher during rapid upward bull runs than during selloffs, unlike equities).
- **Non-Gaussian Error Distributions:** Re-estimate AR-GARCH with a **Student's** t or **Skewed Student's** t distribution to better capture crypto fat tails (excess kurtosis).

C. Expanding the Feature Space (Exogenous Crypto Variables)

- **On-Chain Metrics:** Incorporate blockchain-specific signals, such as active wallet addresses, hash rate, exchange net inflows/outflows, and miner reserve movements.
- **Derivatives & Liquidity Data:** Include perpetual swap **funding rates**, open interest (OI), and liquidations, which serve as direct precursors to overnight price gaps and volatility squeezes.
- **Macro & Sentiment Features:** Integrate the Crypto Fear & Greed Index, US Dollar Index (DXY), and macroeconomic interest rate expectations.

CODE: BITCOIN.ipynb